



Technical Service Bulletin: TSB220076

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Fuel Pump Gear Pump Weep Hole Plugging Procedure to Resolve Nuisance Lubricating Oil and Fuel Leak

Fuel Pump Gear Pump Weep Hole Plugging Procedure to Resolve Nuisance Lubricating Oil and Fuel Leak

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- L9 CM2350 L116B
- L9 CM2350 L119B
- L9 CM2350 L123B
- L9 CM2450 L126B
- L9 CM2450 L169B
- ISX15 CM2250
- ISX15 CM2250 SN
- ISX15 CM2350 X101
- X15 CM2350 X114B
- X15 CM2350 X116B
- X15 CM2350 X123B
- X15 CM2350 X129B
- X15 CM2450 X124B
- X15 CM2450 X134B
- X15 CM2450 X142B
- X15 CM2450 X153B
- X15 CM2450 X154B

Issue Summary

Symptoms:

- Small seepage (normal use) that accumulates over time
- External lubricating oil leak
- External fuel leak

Root Cause:

- Lubricating oil bypassing the fuel pump gear pump shaft seals

Verification

1. Determine if the fuel pump is actively leaking.

- Clean the fuel pump gear pump weep hole with contact cleaner, drying thoroughly.
- Operate engine until operating temperature is reached.
- Inspect the weep hole for leakage.

Note : The weep hole is on the inboard side of the fuel pump gear pump. Use of a mirror or borescope and flashlight may be useful.



Figure 1, Weep Hole Shown Through Handheld Mirror.

- If the fuel pump gear pump weep hole:
 - Continuously leaks; replace the fuel pump gear pump. See corresponding Service Manual. Reference Procedure 005-025 in Section 5.
 - Does **not** leak: fuel pump gear pump is deemed acceptable for weep hole plugging procedure

Resolution

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Plug the fuel pump gear pump weep hole. See Service Instructions section below.

Service Instructions

Optional parts and procedure to plug fuel pump gear pump weep hole:

Parts list:

- 1/4-20 NC set screw or 5/16-18 NC set screw (weep hole plug) – length can vary between .5 in to 1.5 in.
- 1/4-20 NC Tap or 5/16-18 NC
- Loctite® 565 or 577 thread sealant (to improve sealing capability)

See Table 1 for recommended tap for weep hole size.

Table 1, Hole Size to Recommended Tap Size.

Hole Size	Recommended Tap and Plug
.197"	1/4 -20 NC
.225"	1/4 -20 NC
.276"	5/16-18 NC

To plug the fuel pump gear pump weep hole, follow the below instructions:

1. Remove fuel pump gear pump from vehicle.
2. Place fuel pump gear pump in a vice and clean the weep hole region with degreaser, wipe and dry with shop air.



Figure 2, Fuel Pump Gear Pump Weep Hole.

3. Measure the weep hole size with a set of dial calipers or equivalent to determine the recommended tap size according to Table 1.
4. Use the appropriate size tap according to Table 1 to create the threads in the fuel pump gear pump weep hole. No drilling necessary, see Figure 3.

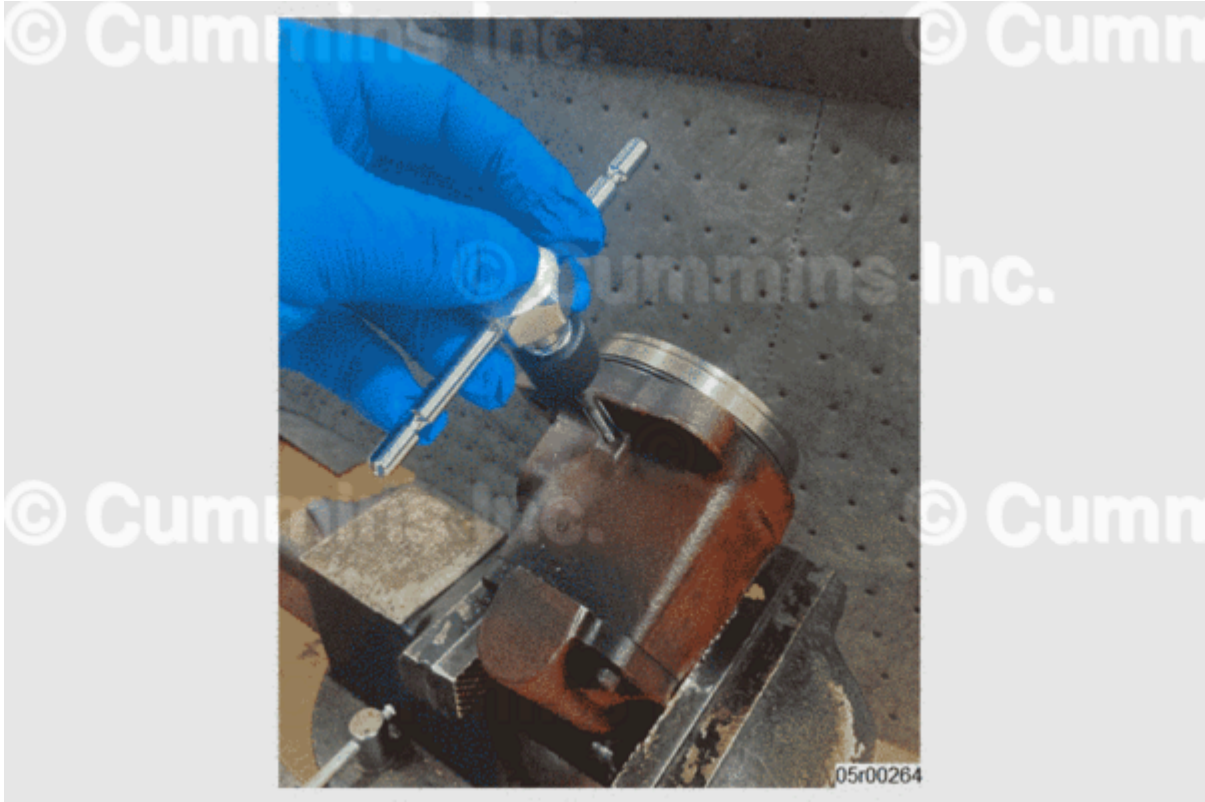


Figure 3, Fuel Pump Gear Pump Weep Hole Tapping.

5. Length of the threads needs to match the length ("L") of the plug as shown on Figure 4.

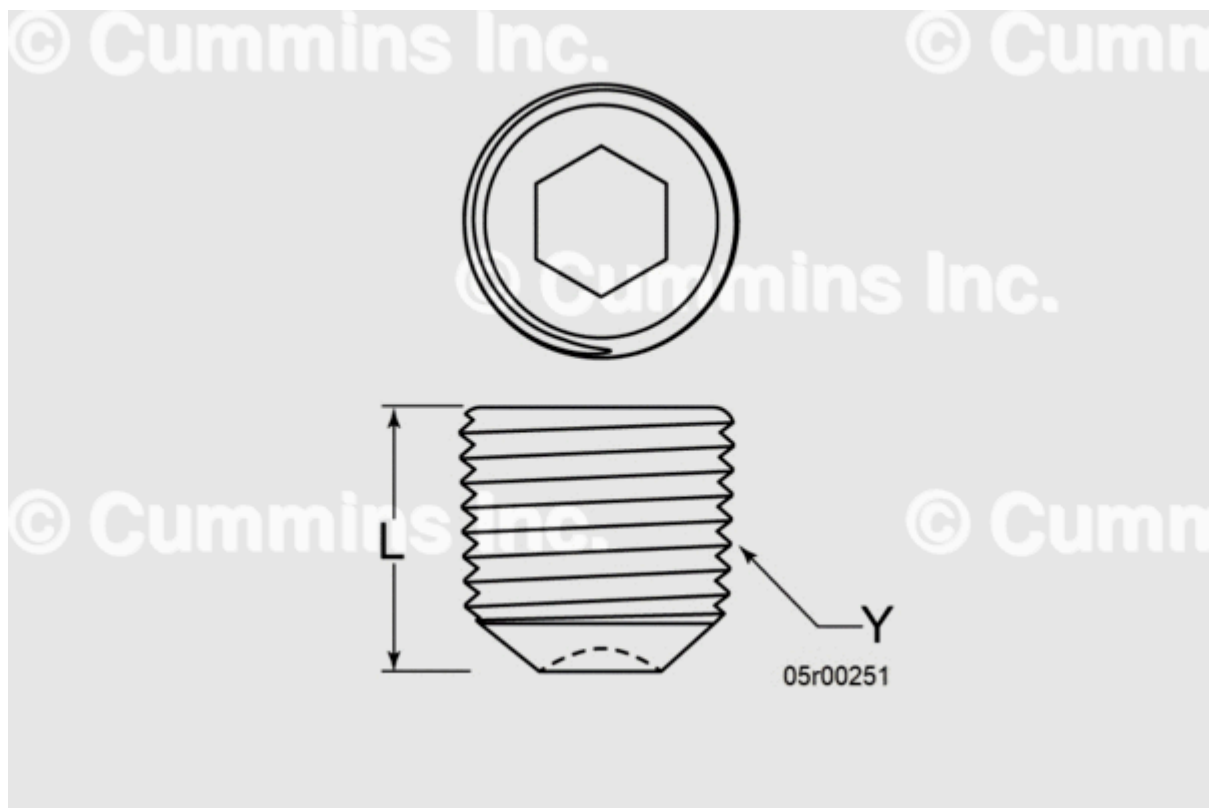


Figure 4, Fuel Pump Gear Pump Weep Hole Threaded Plug Diagram.

6. Use shop air to remove any metal shavings from procedure.



Figure 5, Fuel Pump Gear Pump Weep Hole Threaded Plug Final Assembly.

7. Apply Loctite® 565 or 577 thread sealant on set screw.

8. Install set screw into the threaded weep hole, torque to 103 N•m [76 in-lb]. See final assembly in Figure 5.

9. Install the fuel pump gear pump on vehicle. See corresponding Service Manual. Reference Procedure 005-025 in Section 5.

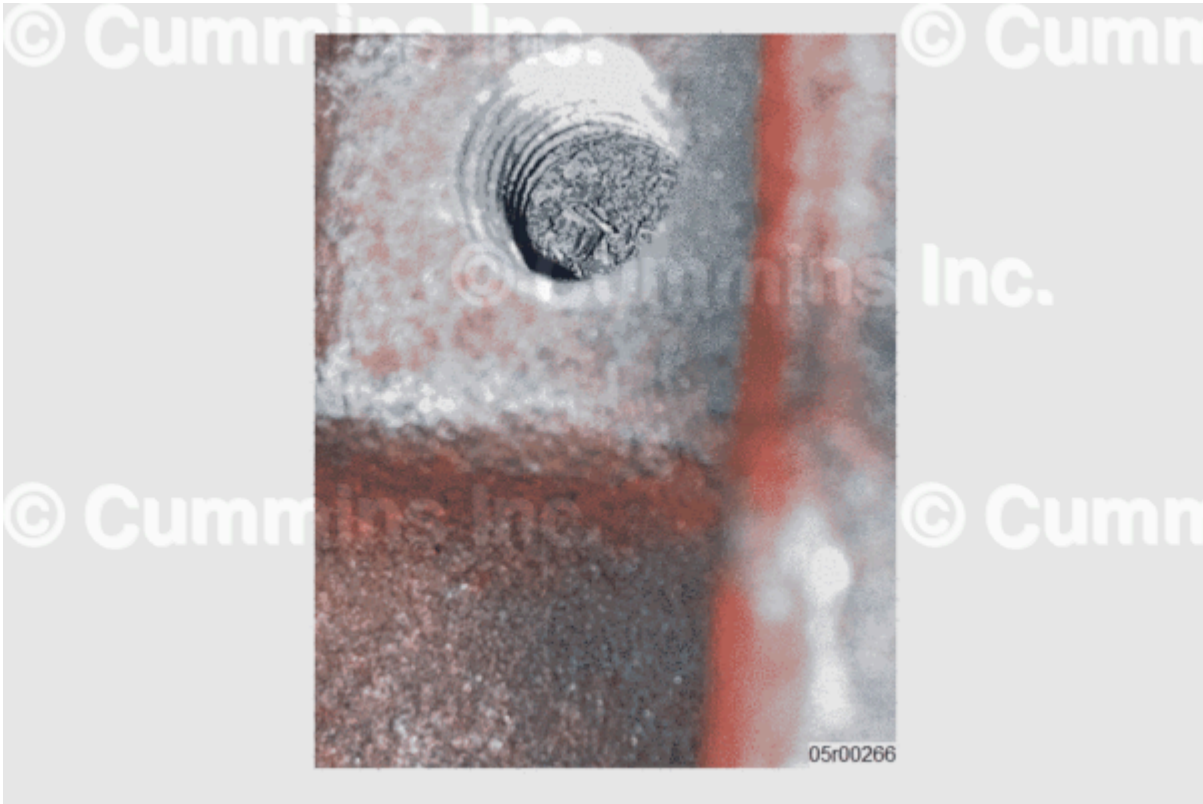


Figure 6, Fuel Pump Gear Pump Filter.

To alleviate concerns on machining debris, the fuel pump gear pump comes equipped with a filter that is located inside the fuel pump gear pump weep hole. See Figure 6. This fuel pump gear pump filter will prevent any debris from reaching fuel pump shaft or seals. Do **not** remove the weep hole filter; leave in place inside the fuel pump gear pump weep hole.

Document History

Date	Details
2022-4-11	Module Created
2022-7-22	Added External lubricating oil leak to Symptoms. Updated Verification section and figures.
2024-9-12	Added Table 1 and updated Figure 5.

